Diagramme de classes AuditFlow

Vue complete de l'application ONCF d'audit interne : objets metier, relations de composition, stockage partage, services, contextes React et ecrans principaux.

Entites metier | Etat partage et persistance | Pages et modules fonctionnels | Contextes et services transverses

BLOC 1 Metier et espace partage

ENTITE
User
- id, name, avatar, email
- firstName, lastName
- role, username, department

ENTITE
AuditorAccount
- heritage de User
- password
- email, username
- department, role

ENTITE
StoredProfile
- avatar, email
- firstName, lastName
- jobTitle, phone
- city, region, country, timezone

ENTITE
AuditMission
- id, title, client, site
- owner, scope, objective
- phase, status, priority
- progress, dueDate
- openFindings, nextAction

ENTITE
MissionTemplate
- id, title, description
- defaultEntity, duration
- focusAreas, deliverables
- kickoffChecklist
- recommendedTeam, useCase

ENTITE
WorkspaceFollowUpAction
- id, title, description
- recommendationNumber
- recommendation, actionDetail
- owner, missionId
- status, evaluation
- startDate, endDate, deadline
- remarks

ETAT PARTAGE
WorkspaceState
- missions: AuditMission[]
- archives: AuditMission[]
- followUpActions: WorkspaceFollowUpAction[]
- studyPhaseState
- orientationReportState
- version

SERVICE
WorkspaceStorageService
- loadWorkspaceState()
- persistWorkspaceState()
- updateWorkspaceState()
- saveWorkspaceMission()
- deleteWorkspaceMission()
- saveWorkspaceFollowUpActions()
- saveWorkspaceStudyPhaseState()
- saveWorkspaceOrientationReportState()

ENUMERATION
Enums principaux
- MissionStatus
- MissionPriority
- WorkspaceFollowUpStatus
- WorkspaceFollowUpEvaluation

Relations clefs
- AuditorAccount herite de User.
- StoredProfile enrichit User via ProfileStorageService.
- WorkspaceState compose missions, archives et followUpActions.
- WorkspaceFollowUpAction pointe optionnellement vers AuditMission via missionId.
- WorkspaceStorageService centralise toutes les mises a jour partagees.
- saveWorkspaceMission synchronise automatiquement missions et archives.
- deleteWorkspaceMission supprime aussi les actions de suivi liees.

BLOC 2

Phase d'etude et rapport

ETAT

StudyState

- knowledge: TabState
- riskListing: TabState
- finalTalk: TabState
- auditPlanning: TabState
- orientationReport: TabState

ETAT

TabState

- documents: DocItem[]
- notes: NoteItem[]
- riskRows: RiskRow[]
- finalTalkRows: FinalTalkRow[]
- auditTests: AuditTestRow[]
- orientationReport
- isGenerated, isValidated
- savedAt, validatedAt, validatedBy

OBJET

CardDef

- id, title, category
- summary, detail

OBJET

DocItem

- id, name, size, addedAt

OBJET

NoteItem

- id, author, content, createdAt

OBJET

RiskRow

- code
- auditableObject
- auditableSubObject
- risk
- operationalControlObjective
- riskAssessment, impact
- probability, priority
- controlMeasure
- decisionStatus

OBJET

FinalTalkRow

- code
- auditableObject
- auditableSubObject
- title
- operationalControlObjective
- riskAssessment, impact
- probability
- controlMeasure / bonnes pratiques

OBJET

AuditTestRow

- testNumber
- controlDescription
- objective, methodology
- auditor, status
- startDate, endDate
- comments, action
- isNonConformity

OBJET

OrientationReportData

- missionLabel, missionReference
- writer, version, reportDate
- missionLead, missionSupervisor
- objective, entity, urgency
- scope, metrics, characteristics
- expectedBenefits, auditeeConcerns
- exclusions
- objectiveRows[]

OBJET

OrientationObjectiveRow

- domain
- objective
- controlCriteria
- risk

ETAT

OrientationPageState

- orientationReport
- isValidated
- savedAt
- validatedAt
- validatedBy

Relations clefs

- StudyState compose cinq TabState.
- TabState compose DocItem, NoteItem, RiskRow, FinalTalkRow et AuditTestRow.
- TabState embarque aussi OrientationReportData.
- RiskRow et FinalTalkRow partagent le meme langage metier du risque.
- OrientationReportData compose plusieurs OrientationObjectiveRow.
- StudyPhasePage persiste son etat dans WorkspaceState.studyPhaseState.
- OrientationReportPage persiste son etat dans WorkspaceState.orientationReportState.

BLOC 3

Services, contextes et hooks

SERVICE

AuthClient

- signUp()
- signInWithPassword()
- resetPassword()
- getUser()
- signOut()

SERVICE

ProfileStorageService

- createDefaultProfile()
- loadStoredProfile()
- saveStoredProfile()
- mergeUserWithStoredProfile()

CONTEXTE

UserContextValue

- user
- error
- isLoading
- checkSession()

CONTEXTE

NotificationsContextValue

- notifications[]
- addNotification()
- clearNotifications()

CONTEXTE

AppLanguageContextValue

- language
- setLanguage()
- usage actuel : francais fixe

HOOK

UseWorkspace

- workspace
- isReady
- ecoute WORKSPACE_UPDATED_EVENT

Relations fonctionnelles

- AuthClient s'appuie sur AuditorAccount pour valider les acces ONCF.
- AuthClient reconstruit l'utilisateur via createStoredUser() puis mergeUserWithStoredProfile().
- UserProvider depend de AuthClient pour charger la session et exposer UserContextValue.
- NotificationsProvider gere un historique local d'AppNotification.
- UseWorkspace recharge WorkspaceState a chaque evenement de mise a jour.
- ProfileStorageService diffuse PROFILE_UPDATED_EVENT pour resynchroniser l'avatar et le profil.
- AppLanguageProvider conserve une interface de langue mais l'application est verrouillee sur fr.

BLOC 4

Pages et navigation

NAVIGATION

Paths

- auth/sign-in, sign-up, reset-password
- dashboard overview
- study-phase, follow-up, rapport
- missions, archives, support
- notifications, account, settings

NAVIGATION

NavItemConfig

- key, title, icon, href
- items[]
- matcher

PAGE

RootPage / SignInForm

- RootPage redirige vers sign-in
- SignInForm appelle AuthClient
- ouvre un panneau de selection des missions
- consomme UseWorkspace et Paths

PAGE

DashboardOverviewPage

- consomme UseWorkspace
- affiche les KPI globaux
- tableau verification des missions
- plein ecran du registre

PAGE

NewMissionPage

- consomme missionTemplates
- genere un AuditMission
- sauvegarde via saveWorkspaceMission()
- notifie la creation

PAGE

MissionsPage / MissionDetailsPage

- lisent workspace.missions
- ouvrent une mission par id
- suppriment via deleteWorkspaceMission()
- emet tent des notifications

PAGE

ArchivesPage

- lit workspace.archives
- ouvre les details mission
- adapte le texte via AppLanguageContext

PAGE

StudyPhasePage

- gere les 5 onglets et leur pagination
- genere, modifie, valide et sauvegarde
- persist StudyState

PAGE

OrientationReportPage

- lit et edite OrientationReportData
- calcule la progression
- persist OrientationPageState

PAGE

FollowUpPage

- utilise workspace.followUpActions
- ajoute, importe, deplace et supprime
- met a jour le diagramme dynamique
- persiste les actions de suivi

PAGE

Pages transverses

- NotificationsPage
- SupportPage
- AccountPage
- SettingsPage

Relations clefs

- Paths alimente la navigation laterale et les liens de pages.
- SignInForm depend de AuthClient et du registre partage des missions.
- DashboardOverviewPage, MissionsPage, MissionDetailsPage et ArchivesPage lisent AuditMission via WorkspaceState.
- NewMissionPage ecrit dans WorkspaceState via saveWorkspaceMission().
- StudyPhasePage ecrit dans studyPhaseState.
- OrientationReportPage ecrit dans orientationReportState.
- FollowUpPage lit et ecrit WorkspaceFollowUpAction[] en temps reel.

ANNEXE

Code Mermaid complet

Cette annexe conserve la version textuelle complete du diagramme de classes, utile pour une regeneration automatique dans un outil Mermaid ou pour une maintenance future dans le depot.

```
classDiagram
direction LR

class User
class AuditorAccount
class StoredProfile
class AppNotification
class AuditMission
class MissionTemplate
class WorkspaceFollowUpAction
class WorkspaceState
class WorkspaceStorageService
class AuthClient
class ProfileStorageService
class UserContextValue
class NotificationsContextValue
class AppLanguageContextValue
class UserProvider
class NotificationsProvider
class AppLanguageProvider
class UseWorkspace
class NavItemConfig
class Paths
class CardDef
class DocItem
class NoteItem
class OrientationObjectiveRow
class OrientationReportData
class OrientationPageState
class FinalTalkRow
class RiskRow
class AuditTestRow
class TabState
class StudyState
class FollowUpDraft
class RootPage
class SignInForm
class DashboardOverviewPage
class MissionsPage
class MissionDetailsPage
class NewMissionPage
class ArchivesPage
class StudyPhasePage
class OrientationReportPage
class FollowUpPage
class NotificationsPage
class SupportPage
class AccountPage
class SettingsPage

AuditorAccount --|> User
WorkspaceState *-- AuditMission
WorkspaceState *-- WorkspaceFollowUpAction
WorkspaceState o-- StudyState
WorkspaceState o-- OrientationPageState
WorkspaceFollowUpAction --> AuditMission
OrientationPageState *-- OrientationReportData
OrientationReportData *-- OrientationObjectiveRow
StudyState *-- TabState
TabState *-- DocItem
TabState *-- NoteItem
TabState *-- RiskRow
TabState *-- FinalTalkRow
TabState *-- AuditTestRow
TabState *-- OrientationReportData
UserProvider --> AuthClient
AuthClient --> AuditorAccount
AuthClient --> ProfileStorageService
ProfileStorageService --> StoredProfile
NotificationsProvider *-- AppNotification
UseWorkspace --> WorkspaceStorageService
SignInForm --> AuthClient
SignInForm --> UseWorkspace
DashboardOverviewPage --> UseWorkspace
MissionsPage --> WorkspaceStorageService
MissionDetailsPage --> WorkspaceStorageService
NewMissionPage --> MissionTemplate
NewMissionPage --> WorkspaceStorageService
ArchivesPage --> UseWorkspace
StudyPhasePage --> StudyState
```

```
StudyPhasePage --> WorkspaceStorageService
OrientationReportPage --> OrientationPageState
OrientationReportPage --> WorkspaceStorageService
FollowUpPage --> WorkspaceFollowUpAction
FollowUpPage --> WorkspaceStorageService
NavItemConfig --> Paths
```